

**STUDENT NAME: HONEY**

**ROLL NUMBER: 25BSCS007**

**SECTION: A**

**SUBJECT NAME: OBJECT ORIENTED PROGRAMMING**

**LAB TITLE: ATM SYSTEM**

**Lab title :** Multi-Feature Secure ATM System Using Java

**Problem Statement:** Develop a Java program that simulates a secure ATM system supporting multiple users with unique accounts and PINs. The system must provide functionalities including: secure PIN verification (maximum three attempts), deposit, withdrawal with balance and daily limit checks, fund transfers between accounts, balance inquiry, and a mini statement to track transactions. Users should also be able to change their PIN and view a session summary with totals and final balances. The program should validate all inputs, prevent invalid operations such as negative amounts or exceeding limits, and ensure a smooth, repeating menu-driven interaction. This lab assesses the student's understanding of variables, data types, loops, conditionals, arrays, input handling, counters, and output formatting in Java.

### **Objectives:**

- To implement basic Java constructs: variables, data types, operators, loops, conditionals, and arrays.
- To design a menu-driven application with proper input validation and user interaction.
- To implement secure PIN verification with limited login attempts.
- To handle deposit, withdrawal, and transfer operations with balance and daily limits.
- To maintain a mini statement using arrays to track transactions.
- To provide a change PIN feature and final session summary for the user.
- To develop problem-solving skills and reinforce step-by-step testing for a large, real-world application.

**Tools / Requirements:** Programming Language : Java, IDE : Notepad

### **Source code:**

```
import java.util.*;
```

```
public class ATMLAB2OOP3 {
```

```

public static void main(String args[]) {

    Scanner s = new Scanner(System.in);
    int[] accountNs = {10001, 10020, 10300, 14000, 10500};
    int[] pins = new int[5];
    double[] totalAmount = new double[5];
    double[] dailyWdL = {100000, 100000, 100000, 100000, 100000};
    double[] dailyTfL = {200000, 200000, 100000, 100000, 100000};

    while(true){
        System.out.println("Enter your account number: ");
        int acNo = s.nextInt();

        int curU = -1 ;
        for(int j = 0; j < accountNs.length; j++) {
            if(accountNs[j] == acNo){
                curU = j;
                break; }}

        if(curU == -1) {
            System.out.println("Account does not exist, Try again");
            continue; }

        System.out.println("Correct account number!, Current user index: " + curU);

        if( pins[curU] == 0){
            System.out.println("Set your pin: ");
            pins[curU] = s.nextInt();
            System.out.println("Pin has been set successfully!");}

        int aP = 0;
        boolean logIn = false;
        while (aP < 3) {
            System.out.print("Enter your Pin to login: ");
            int userP = s.nextInt();
            if (userP == pins[curU]) {
                logIn = true;
                break; }
            else {
                aP++;
                System.out.println("Wrong Pin, Try again "); }}

        if (!logIn) {
            System.out.println("3 wrong Pins. Account blocked!");
            continue; }
    }
}

```

```

System.out.println("Pin correct! You are logged in.");

int choice;

do {
    System.out.println("=====MENU SYSTEM=====");
    System.out.println("1. Deposit Money");
    System.out.println("2. Withdraw Money");
    System.out.println("3. Transfer Money");
    System.out.println("4. Check Balance");
    System.out.println("0. Exit");
    System.out.print("Enter your choice: ");
    choice = s.nextInt();

    switch(choice) {
        case 1: System.out.print("Enter amount to deposit: ");
            double depositAmount = s.nextDouble();
            if(depositAmount > 0 && depositAmount <= dailyTfL[curU]) {
                totalAmount[curU] += depositAmount ;
                dailyTfL[curU] -= depositAmount;
                System.out.println("Deposit successful! Your current balance: " + totalAmount[curU]);
                System.out.println("Remaining daily deposit limit: " + dailyTfL[curU]);}
            else {
                System.out.println("Invalid amount or exceeds daily limit!");}
            break;

        case 2: System.out.println("Enter amount to withdraw: ");
            double withdrawAmount = s.nextDouble();
            if (withdrawAmount > 0 && withdrawAmount <= totalAmount[curU] &&
withdrawAmount <= dailyWdL[curU]) {
                totalAmount[curU] -= withdrawAmount;
                dailyWdL[curU] -= withdrawAmount;
                System.out.println("Withdrawal successful! Your current balance: " +
totalAmount[curU]);
                System.out.println("Remaining daily withdrawal limit: " + dailyWdL[curU]); }
            else { System.out.println("Invalid amount, insufficient balance, or exceeds daily
limit!");}
            break;

        case 3: System.out.println("Enter recipient account number: ");
            int recAcc = s.nextInt();
            int recIndex = -1;
            for (int j = 0; j < accountNs.length; j++) {
                if (accountNs[j] == recAcc) {
                    recIndex = j;
                    break;}}
    }
}

```

```

        if (recIndex == -1) {
            System.out.println("Recipient account not found!");
            break;}
        System.out.print("Enter amount to transfer: ");
        double transferAmount = s.nextDouble();
        if (transferAmount > 0 && transferAmount <= totalAmount[curU] &&
transferAmount <= dailyTfL[curU]) {
            totalAmount[curU] -= transferAmount;
            totalAmount[recIndex] += transferAmount;
            dailyTfL[curU] -= transferAmount;
            System.out.println("Transfer successful! Your current balance: " +
totalAmount[curU]);
            System.out.println("Remaining daily transfer limit: " + dailyTfL[curU]);}

        else {System.out.println("Invalid amount or insufficient balance!");}
        break;
    case 4:  System.out.println("Your current balance: " + totalAmount[curU]);
        break;

    case 0: System.out.println("Exiting menu. Thank you!");
        break;
    default:
        System.out.println("Invalid choice!");} }
while (choice != 0);} } }

```

## Output Screenshot:

```

D:\HONEY TASKS 007\2ND SEMESTER\OOP LABS> javac ATMLAB200P3.java
D:\HONEY TASKS 007\2ND SEMESTER\OOP LABS> java ATMLAB200P3
Enter your account number:
10001
Correct account number!, Current user index: 0
Set your pin:
1789
Pin has been set successfully!
Enter your Pin to login: 1789
Pin correct! You are logged in.
=====MENU SYSTEM=====
1. Deposit Money
2. Withdraw Money
3. Transfer Money
4. Check Balance
0. Exit
Enter your choice: 1
Enter amount to deposit: 120000
Deposit successful! Your current balance: 120000.0
Remaining daily deposit limit: 80000.0
=====MENU SYSTEM=====
1. Deposit Money
2. Withdraw Money
3. Transfer Money
4. Check Balance
0. Exit
Enter your choice: 3
Enter recipient account number:
10020

```

```
4. Check Balance
0. Exit
Enter your choice: 3
Enter recipient account number:
10020
Enter amount to transfer: 1234
Transfer successful! Your current balance: 118766.0
Remaining daily transfer limit: 78766.0
=====MENU SYSTEM=====
1. Deposit Money
2. Withdraw Money
3. Transfer Money
4. Check Balance
0. Exit
Enter your choice: 4
Your current balance: 118766.0
=====MENU SYSTEM=====
1. Deposit Money
2. Withdraw Money
3. Transfer Money
4. Check Balance
0. Exit
Enter your choice: 0
Exiting menu. Thank you!
Enter your account number:
10020
Correct account number!, Current user index: 1
Set your pin:
0987
Pin has been set successfully!
```

```
0987
Pin has been set successfully!
Enter your Pin to login: 0987
Pin correct! You are logged in.
=====MENU SYSTEM=====
1. Deposit Money
2. Withdraw Money
3. Transfer Money
4. Check Balance
0. Exit
Enter your choice: 4
Your current balance: 1234.0
=====MENU SYSTEM=====
1. Deposit Money
2. Withdraw Money
3. Transfer Money
4. Check Balance
0. Exit
Enter your choice: 2
Enter amount to withdraw:
1000
Withdrawal successful! Your current balance: 234.0
Remaining daily withdrawal limit: 99000.0
=====MENU SYSTEM=====
1. Deposit Money
2. Withdraw Money
3. Transfer Money
4. Check Balance
0. Exit
Enter your choice: |
```